Sheth L.U.J. And Sir M.V. College

Aim:Wireless Sensor Network Deployment and Coverage Analysis

Simulate random and grid deployment of sensor nodes and evaluate network coverage and connectivity.

CODE:

```
# ================================================
# WSN Grid Deployment and Coverage Analysis
# NS2.35 Compatible
# ================================================

set ns [new Simulator]

# Trace Files
set tracefile [open wsn.tr w]
$ns trace-all $tracefile

set namfile [open wsn.nam w]
$ns namtrace-all-wireless $namfile 500 500

# Topography
set topo [new Topography]
$topo load_flatgrid 500 500

# Simulation Parameters
set val(chan)      Channel/WirelessChannel
set val(prop)      Propagation/TwoRayGround
set val(netif)     Phy/WirelessPhy
set val(mac)       Mac/802_11
set val(ifq)       Queue/DropTail/PriQueue
set val(ll)        LL
set val(ant)       Antenna/OmniAntenna
set val(ifqlen)    50
set val(nn)        9
set val(rp)        AODV

create-god $val(nn)
```

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```
# Configure Nodes
$ns node-config \
    -adhocRouting $val(rp) \
    -llType $val(ll) \
    -macType $val(mac) \
    -ifqType $val(ifq) \
    -ifqLen $val(ifqlen) \
    -antType $val(ant) \
    -propType $val(prop) \
    -phyType $val(netif) \
    -channelType $val(chan) \
    -topoInstance $topo \
    -agentTrace ON \
    -routerTrace ON \
    -macTrace ON

# Create Nodes
for {set i 0} {$i < $val(nn)} {incr i} {
    set node(i)[ns node]
}

# ================================================
# GRID DEPLOYMENT (3x3)
# ================================================

$node(0) set X_ 100
$node(0) set Y_ 100

$node(1) set X_ 200
$node(1) set Y_ 100

$node(2) set X_ 300
$node(2) set Y_ 100

$node(3) set X_ 100
$node(3) set Y_ 200

$node(4) set X_ 200
$node(4) set Y_ 200

$node(5) set X_ 300
```

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```
$node(5) set Y_ 200

$node(6) set X_ 100
$node(6) set Y_ 300

$node(7) set X_ 200
$node(7) set Y_ 300

$node(8) set X_ 300
$node(8) set Y_ 300

for {set i 0} {$i < $val(nn)} {incr i} {
    node(i) set Z_ 0
    $ns initial_node_pos node(i) 20
}

# ================================================
# Traffic
# ================================================

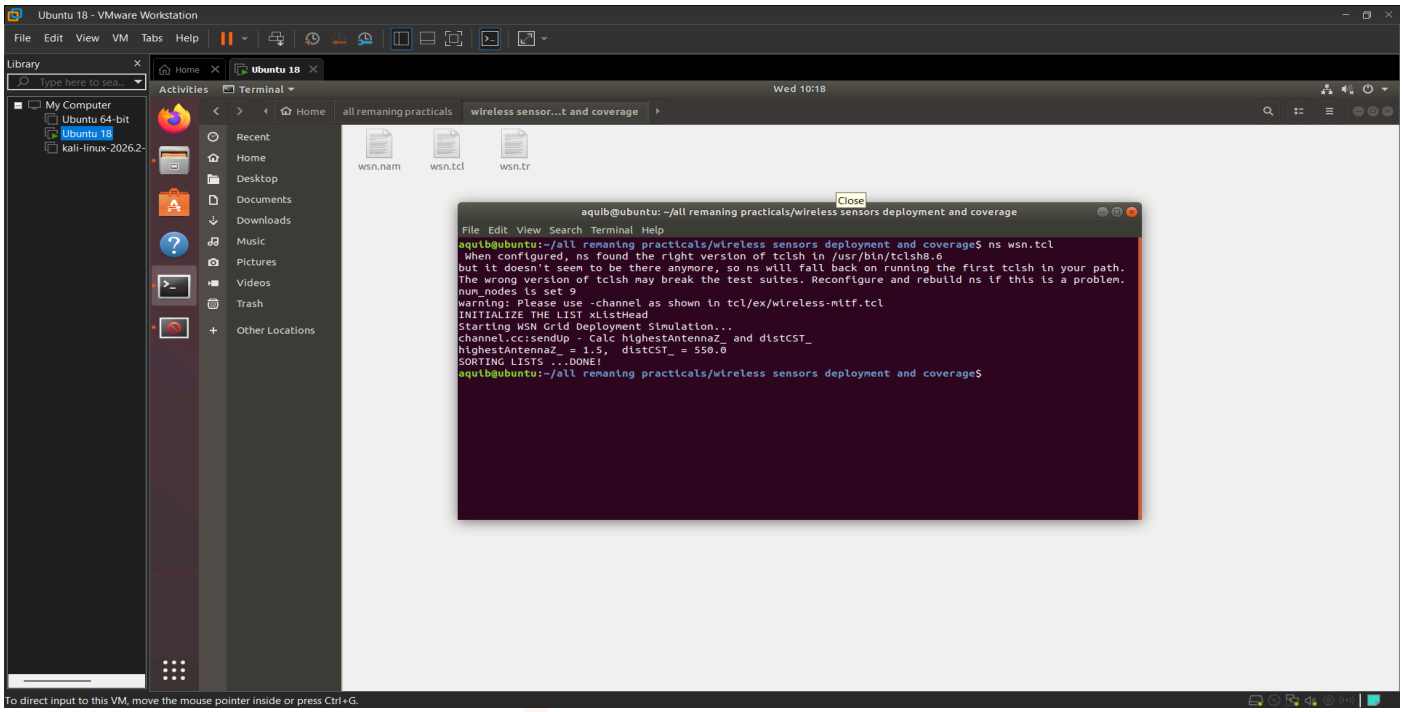
set udp0 [new Agent/UDP]
$ns attach-agent $node(0) $udp0
set null0 [new Agent/Null]
$ns attach-agent $node(8) $null0
$ns connect $udp0 $null0
set cbr0 [new Application/Traffic/CBR]
$cbr0 set packetSize_ 512
$cbr0 set interval_ 0.1
$cbr0 attach-agent $udp0
nsat1.0"cbr0 start"
nsat8.0"cbr0 stop
# ================================================
# Finish Procedure
# ================================================
proc finish {} {
    global ns tracefile namfile
    $ns flush-trace
    close $tracefile
    close $namfile
    exec nam wsn.nam &
    exit 0
```

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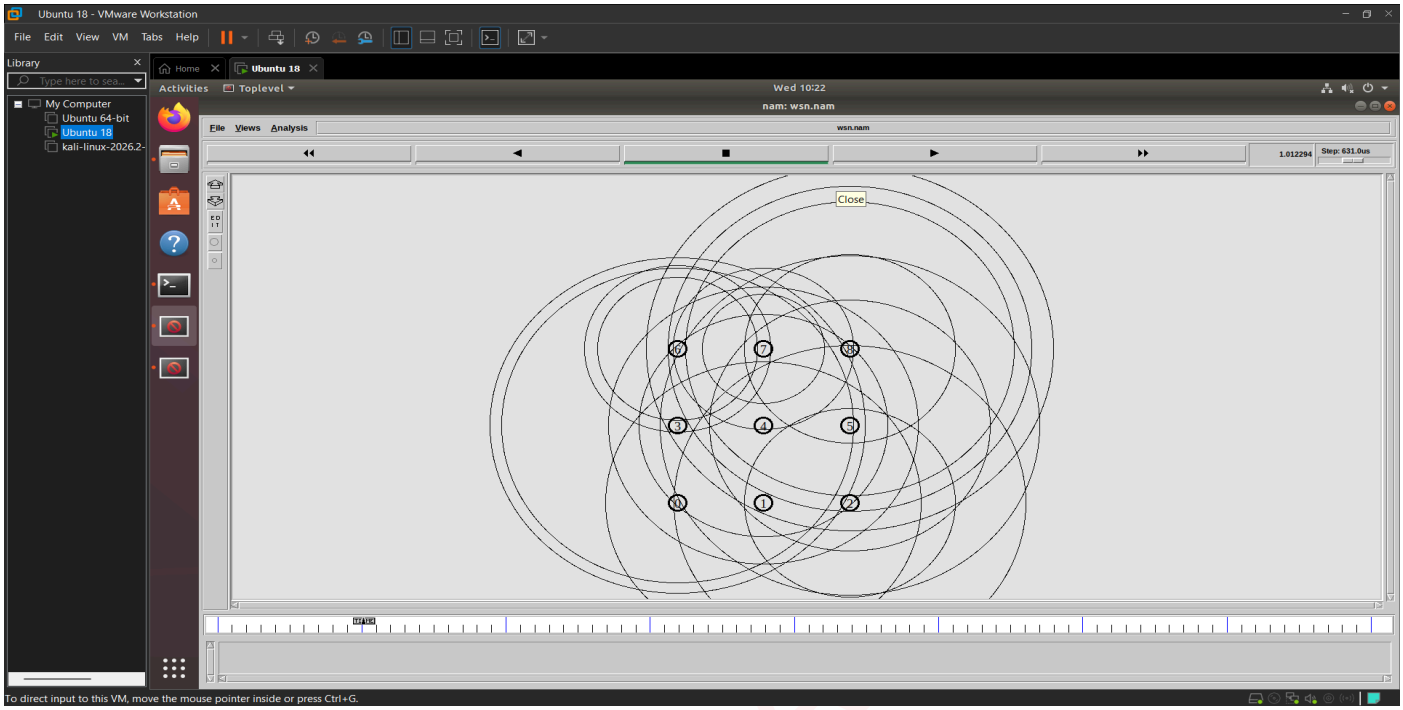
```
}
$ns at 10.0 "finish"
puts "Starting WSN Grid Deployment Simulation..."
$ns run
```

OUTPUT:



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Result

The WSN grid deployment simulation was successfully executed. Sensor nodes were uniformly deployed, and data packets were transmitted successfully from the source node to the destination node.

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